Examiner only

6. A teacher demonstrates heat transfer by infra-red radiation.

Experiment 1

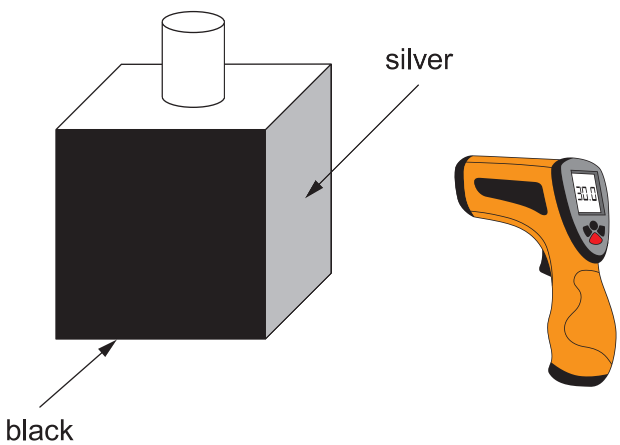
The teacher places a heater in between a black surface and a silver surface.

After a few minutes the temperature of each surface is recorded using an infra-red thermometer.

Experiment 2

The teacher fills the container with boiling water.

The infra-red thermometer is then used to take readings next to the side of the container which is black and the side which is silver.



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The following results were obtained.

	Temperature (°C)	
	Black surface	Silver surface
Experiment 1	46	32

	Temperature (°C)	
	Black surface	Silver surface
Experiment 2	90	65

(a) Use the data to complete the following sentences.

(i) The .... surface is the better emitter of heat

as shown by the results for **Experiment** .... because ....

.... [2]

(ii) The .... surface is the better absorber of heat

as shown by the results for **Experiment** .... because ....

.... [2]

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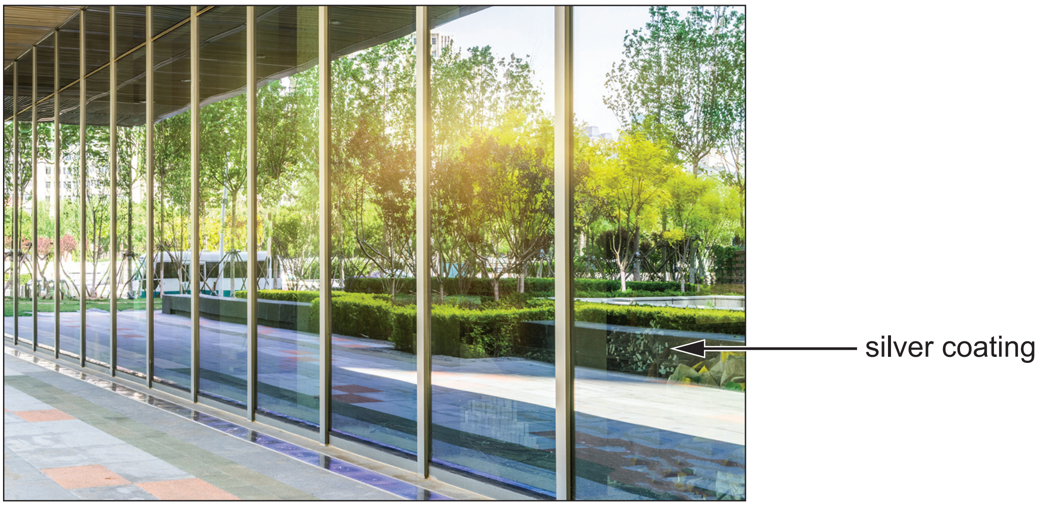
(b) Silver coating can be put on the outside of windows to help to keep houses cool in summer and warm in winter.



State why the silver coating helps to keep the house warm in the winter. [1]

....

....

(c) A family consider different methods of reducing their energy bills.
Data about these methods are given in the table below.

Method	Cost (£)	Yearly savings (£)
Cavity wall insulation	640	160
Loft insulation	480	160
Silver coating	500	25

(i) Calculate the payback time for the silver coating. [2]

payback time = ... years

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(ii) The family can only choose to fit one of the items.
Ieuan suggests that it is better to fit cavity wall insulation than loft insulation.
Use data from the table to explain whether you agree. [2]
Space for calculations.

....

....

....

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